

# Optimal bidding and scheduling of AA-CAES based energy hub considering cascaded consumption of heat

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## ABSTRACT

The increasing level of uncertainty caused by high penetration of renewable energy and the widening gap of peak-valley demands call for the deployment of energy storage in power systems. Advanced-adiabatic compressed air energy storage (AA-CAES) is a large-scale physical energy storage technology with the merits of long lifetime, low environmental impact, and no emission. Moreover, AA-CAES works with electricity and heat, making it an excellent choice to realize the framework of energy hub. This paper proposes a cogeneration and storage architecture of an AA-CAES based energy hub in an industrial park. Particularly, the cascaded use of thermal energy to supply heat demands with different temperatures is modeled. A bi-level optimization model is established to study the optimal bidding and scheduling of AA-CAES based energy hub in the day-ahead market. The upper level is the power purchase and self-scheduling of the energy hub, aiming at minimizing the daily operation cost; the lower level represents the market clearing problem which determines the electricity price based on alternating-current optimal power flow. A radial basis function based surrogate optimization method is developed to solve the bi-level model with a nonlinear lower level problem. The problem is decomposed into second-order cone programs and a nonlinear surrogate model which can be solved without much computational effort. Numerical examples verify the effectiveness of the proposed method.

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## 1. Introduction

Electricity and heat are two main forms of energy consumption. Cogeneration can make use of waste heat during electricity generation and thus improve overall energy efficiency, so the integrated power and heat system has received wide attention [1]. On the other hand, the development of renewable energy and the increased gap between peak and valley demands require flexible resources for uncertainty mitigation and peak regulation. Energy storage is a main sort of flexibility resource, but the cost of electrical energy storage is still relatively high [2]. In the heat-power integrated energy system, the excessive electricity can be converted into heat that is convenient for storage, which further improves operating flexibility. As the core of the heat-power integrated energy system, energy hub (EH) plays the role of co-generation and storage. In the existing work, an energy hub is mainly composed of independent components such as combined-heat-and-power unit,

battery energy storage, and thermal storage tanks, etc. [3,4]. Advanced-adiabatic compressed air energy storage (AA-CAES) is a large-scale clean physical energy storage technology that naturally possesses co-generation and storage abilities, making it an ideal realization of energy hub [5,6].

Traditional work on modeling AA-CAES mainly focuses on the simulation and analysis of thermodynamics [7,8]. A thermodynamic simulation model of AA-CAES system based on static and dynamic processes of air storage is built in Ref. [9], where the dynamic compressed air storage model demonstrates the influence of time-varying temperature and pressure of air on performance indicators like round-trip efficiency. AA-CAES configuration optimization and efficiency improvement are studied in Refs. [10,11]. Ref. [10] presents a thermodynamic model with low temperature thermal energy storage (TES) integration, and ref. [11] proposes a novel system with variable pressure ratio based on organic Rankine cycle. Refs. [12,13] conduct detailed thermodynamic analysis of multi-stage AA-CAES system. In Ref. [12], the influence of design parameters on AA-CAES system efficiency is analyzed and the comprehensive design standard of AA-CAES is studied. Energy and

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**Nomenclatures***Indices and sets*

|                  |                                    |
|------------------|------------------------------------|
| $i, j$           | Index of buses                     |
| $l$              | Index of lines                     |
| $t$              | Index of time periods              |
| $\pi(i)$         | Set of downstream buses of bus $i$ |
| $(\cdot)_i$      | Symbol associated with bus $i$     |
| $(\cdot)_{ij}^l$ | Symbol associated with line $l$    |

*Parameters*

|                       |                                                        |
|-----------------------|--------------------------------------------------------|
| $c_p^a/c_{oil}$       | Specific heat capacity of air/heat transfer fluid      |
| $V$                   | Volume of air storage tank                             |
| $R_g$                 | Gas constant of air                                    |
| $\tau^{st}$           | Temperature of air storage tank                        |
| $\eta_i^c$            | Efficiency of $i$ th stage compressor                  |
| $\eta_j^e$            | Efficiency of $j$ th stage turbine                     |
| $\eta^H/\eta^L$       | Efficiency of electric heater/heat pump                |
| $\eta_i^{c,ex}$       | Efficiency of $i$ th compression stage heat exchanger  |
| $\eta_j^{e,ex}$       | Efficiency of $j$ th expansion stage heat exchanger    |
| $S^U$                 | Startup cost of compressor/turbine                     |
| $k_1$                 | Penalty of unit high-temperature load shedding         |
| $k_2$                 | Penalty of unit low-temperature load shedding          |
| $k_3$                 | Penalty of unit electric load shedding                 |
| $a_j$                 | Second-order generator cost coefficient at node $j$    |
| $b_j$                 | First-order generator cost coefficient at node $j$     |
| $\mu$                 | Electricity price at slack bus                         |
| $P_i^c/P_j^e$         | Rated power of $i$ th/ $j$ th compressor/turbine       |
| $m^{H,u}$             | Upper bound of fluid mass in high-temperature tank     |
| $m^{H,l}$             | Lower bound of fluid mass in high-temperature tank     |
| $m^{L,u}$             | Upper bound of fluid mass in low-temperature tank      |
| $m^{L,l}$             | Lower bound of fluid mass in low-temperature tank      |
| $m^{O,u}$             | Upper bound of fluid mass in ordinary-temperature tank |
| $m^{O,l}$             | Lower bound of fluid mass in ordinary-temperature tank |
| $n_c/n_e$             | Number of compression/expansion stage                  |
| $pr^{st,l}$           | Lower bound of pressure in air storage tank            |
| $pr^{st,u}$           | Upper bound of pressure in air storage tank            |
| $\beta_i^c$           | Rated compression ratio of $i$ th compressor           |
| $\beta_j^e$           | Rated expansion ratio of $j$ th turbine                |
| $r_{ij}^l$            | Resistance on line $ij$                                |
| $x_{ij}^l$            | Reactance on line $ij$                                 |
| $z_{ij}^l$            | Impedance on line $ij$                                 |
| $\dot{m}^c/\dot{m}^e$ | Rated mass flow rate during air compression/expansion  |
| $\dot{m}^{c,l}$       | Lower bound of air mass flow rate during compression   |
| $\dot{m}^{e,l}$       | Lower bound of air mass flow rate during expansion     |
| $\dot{m}^{c,u}$       | Upper bound of air mass flow rate during compression   |
| $\dot{m}^{e,u}$       | Upper bound of air mass flow rate during expansion     |

*Variables*

|                     |                                               |
|---------------------|-----------------------------------------------|
| $p_t^{ch}/p_t^{ds}$ | Total charging/discharging power              |
| $P_{i,t}^c$         | Power consumption of $i$ th stage compression |

|                             |                                                                        |
|-----------------------------|------------------------------------------------------------------------|
| $P_{j,t}^e$                 | Power generation of $j$ th stage expansion                             |
| $\dot{m}_t^c/\dot{m}_t^e$   | Air mass flow rate during compression/expansion                        |
| $u_t^c/u_t^e$               | Binary variable to indicate the charging/discharging state             |
| $pr_t^{st}$                 | Pressure of air storage tank                                           |
| $\tau_{i,t}^{c,in}$         | Inlet air temperature of $i$ th stage compressor                       |
| $\tau_{i,t}^{c,out}$        | Outlet air temperature of $i$ th stage compressor                      |
| $\tau_{a,n_c,t}^{c,ex,out}$ | Outlet air temperature of $n_c$ th compression stage heat exchanger    |
| $\tau_{j,t}^{e,in}$         | Inlet air temperature of $j$ th stage turbine                          |
| $\tau_{j,t}^{e,out}$        | Outlet air temperature of $j$ th stage turbine                         |
| $\tau_{i,t}^{c,ex,in}$      | Inlet fluid temperature of $i$ th stage compression                    |
| $\tau_{i,t}^{c,ex,out}$     | Outlet fluid temperature of $i$ th stage compression                   |
| $\tau_{j,t}^{e,ex,in}$      | Inlet fluid temperature of $j$ th stage expansion                      |
| $\tau_{j,t}^{e,ex,out}$     | Outlet fluid temperature of $j$ th stage expansion                     |
| $m_t^H$                     | Total fluid mass in high-temperature tank                              |
| $m_t^L$                     | Total fluid mass in low-temperature tank                               |
| $m_t^O$                     | Total fluid mass in ordinary-temperature tank                          |
| $\dot{m}_t^{Hin,c}$         | Inflow rate of high-temperature tank from compression                  |
| $\dot{m}_t^{Oin,e}$         | Inflow rate of ordinary-temperature tank from expansion                |
| $\dot{m}_t^{Hout,e}$        | Outflow rate of high-temperature tank for expansion                    |
| $\dot{m}_t^{Oout,c}$        | Outflow rate of ordinary-temperature tank for compression              |
| $\dot{m}_t^{Hin,c_i}$       | Inflow rate of high-temperature tank from $i$ th stage compression     |
| $\dot{m}_t^{Hout,e_j}$      | Outflow rate of high-temperature tank for $j$ th stage expansion       |
| $\dot{m}_t^{Oout,c_i}$      | Outflow rate of ordinary-temperature tank for $i$ th stage compression |
| $\dot{m}_t^{Oin,e_j}$       | Inflow rate of ordinary-temperature tank from $j$ th stage expansion   |
| $\dot{m}_t^{Hin,O}$         | Ordinary-temperature inflow rate of high-temperature tank              |
| $\dot{m}_t^{Lin,O}$         | Ordinary-temperature inflow rate of low-temperature tank               |
| $\dot{m}_t^{Hout,H}$        | High-temperature outflow rate for high-temperature load                |
| $\dot{m}_t^{Hout,L}$        | High-temperature outflow rate for low-temperature load                 |
| $\dot{m}_t^{Lout,L}$        | Low-temperature outflow rate for low-temperature load                  |
| $\dot{m}_t^{Oin,HH}$        | High-temperature outflow rate of high-temperature load exchanger       |
| $\dot{m}_t^{Oin,HL}$        | High-temperature outflow rate of low-temperature load exchanger        |
| $\dot{m}_t^{Oin,LL}$        | Low-temperature outflow rate of low-temperature load exchanger         |
| $P_t^H/P_t^L$               | Power consumption of electric heater/heat pump                         |
| $\Delta\tau^H$              | Difference between high temperature and ordinary temperature           |
| $\Delta\tau^L$              | Difference between low temperature and ordinary temperature            |

|                    |                                                                                         |                     |                                                          |
|--------------------|-----------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------|
| $\tau^H$           | Temperature of high-temperature tank                                                    | $h_t^{Lde}$         | Low-temperature load shedding                            |
| $\tau^L$           | Temperature of low-temperature tank                                                     | $p_t^{de}$          | Electric load shedding                                   |
| $\tau^O$           | Temperature of ordinary-temperature tank                                                | $h_t^{Hd}$          | High-temperature load                                    |
| $z_1$              | Binary variable to decide cross-grade heating state                                     | $h_t^{Ld}$          | Low-temperature load                                     |
| $SU_\xi^c$         | Startup cost of compressors                                                             | $p_t^{fd}$          | Electric load                                            |
| $SU_\xi^e$         | Startup cost of turbines                                                                | $p_t^b$             | Power purchase                                           |
| $h_t^{Hdsc}$       | Thermal energy supplied for high-temperature load                                       | $\lambda_t$         | Nodal price of AA-CAES connected node                    |
| $h_t^{Ldsc}$       | Thermal energy supplied for low-temperature load                                        | $p_{ij}^l/Q_{ij}^l$ | Active/reactive power flow on line $ij$                  |
| $p_t^{dsc}$        | Electric power supplied for electric load                                               | $p_j^g/q_j^g$       | Active/reactive power of generator at node $j$           |
| $\Delta\tau^{Hd}$  | Temperature gap of high-temperature fluid before and after high-temperature load supply | $I_{ij}^l$          | Square of current on line $ij$                           |
| $\Delta\tau^{HLd}$ | Temperature gap of high-temperature fluid before and after low-temperature load supply  | $p_j^d/q_j^d$       | Active/reactive power demand at node $j$                 |
| $\Delta\tau^{Ld}$  | Temperature gap of low-temperature fluid before and after low-temperature load supply   | $U_i/U_j$           | Square of voltage at node $ij$                           |
| $h_t^{Hde}$        | High-temperature load shedding                                                          | $p_{0j}^l$          | Active power flow on line between slack bus and node $j$ |

exergy analysis of AA-CAES system is discussed in Refs. [14,15]; results indicate that a possible way to improve efficiency and reduce exergy destruction is to divide compression and expansion process into more steps and using more heat exchangers between steps. An experimental study of a pilot AA-CAES is conducted where TES uses water as thermal energy storage working medium in Ref. [16]. In addition to steady state thermodynamic descriptions, dynamic modeling of AA-CAES is studied as well. For example, ref. [17] proposes a dynamic system model with economic analysis for MW-scale AA-CAES and applies the model in micro grid where AA-CAES serves as emergency backup power source. Ref. [18] establishes a comprehensive dynamic model of supercritical CAES system to explore transient characteristics and control methodology, where important factors including volume effect and thermal inertia are considered. Ref. [19] develops a simulation software tool considering complex dynamic characteristics caused by multi-physical processes in AA-CAES. A certain number of validated simulation models about pneumatics, thermodynamics, heat transfer, electrical machines and power grids are included in the developed tool. Ref. [20] applies a differential-algebraic equation constrained optimization to model a state-space set-point control of AA-CAES under power tracking mode. The above works have provided detailed modeling and description of all aspects inside AA-CAES. Nonetheless, these sophisticated models are too complicated to be used in the operational study of the integrated energy system, because they are usually described by nonlinear differential-algebraic equations which is computationally challenging.

Some recent work considers AA-CAES in power system dispatch. Ref. [21] formulates a bi-objective dynamic optimal power flow problem taking AA-CAES into account; the reactive power support capability AA-CAES is explored. In Ref. [22], a tri-state model of AA-CAES considering its part load characteristics is proposed and an efficient linearization method is applied. Ref. [23] raises a bi-level optimal dispatch strategy for realizing the balance between supply and demand in regional micro energy system with AA-CAES, where the upper level problem calculates the optimal initial storage volume, and the lower level optimizes the hourly operation strategy considering the thermo-economic characteristics. Scheduling of regional integrated energy system considering heat and reserve provided by AA-CAES is discussed in Ref. [24]. In the scheduling framework, the multi-type applications of AA-CAES and trading strategy of energy hub operator in both external market and internal micro-market are considered. A joint energy and

reserve optimization scheduling model is investigated for the power system with AA-CAES in Ref. [25]; the scheduling model takes the limits on the reserve capacities of thermal power units and interruptible loads caused by AA-CAES into account. An optimal dispatch model of zero-carbon-emission micro Energy Internet integrated with AA-CAES system is presented in Ref. [26], for the purposes of reducing curtailed wind power and operation cost compared with traditional networks. An optimization model is developed in Ref. [27] to determine the performance of a hybrid power system which includes hydro, thermal, wind and solar powered units along with CAES facilities. Ref. [28] proposes a game theory-based multi-agent capacity optimization method for an integrated energy system (IES) with CAES to analyze the benefit interactions. The results show that the access of CAES could improve economic performance of the IES and the completely cooperative game model yields better economic performance for the whole and for individuals. Research related to optimal bidding of CAES in electricity market is conducted in Refs. [29–31]. Ref. [29] proposes a hybrid robust-stochastic based mixed-integer linear programming approach for optimal bidding and offering strategies of CAES (compressed air energy storage), where the uncertainty of market price is considered with sets of scenarios via stochastic method and uncertainty of maximum capacity of cavern is through robust method. A robust optimization approach is employed in Ref. [30] to achieve the offering and bidding curves of CAES which should be submitted to the pool market. And a two-stage stochastic programming based method is proposed in Ref. [31] to solve the optimal bidding of a generating company owning CAES, wind and thermal units in energy and reserve markets. In the aforementioned research, the external energy carrier is electricity, and thermal energy conversion only happens inside the AA-CAES.

The cogeneration and storage potential of AA-CAES has also been noticed. An MILP-based combined heat and power dispatch model of AA-CAES is established in Ref. [32]. AA-CAES can share its thermal storage tank with electricity and heating system to increase the flexibility of the integrated energy system. Ref. [33] proposes a new structure for an advanced trigenerative CAES system, where a burner is added before the turbine to heat inlet air. Three operation strategies of turbine are analyzed to control output power of system under off-design conditions. Such a structure is also studied in Refs. [34–36]. Ref. [34] derives the guideline for designing the trigenerative CAES based on an accurate thermodynamic model and parametric optimization; ref. [35] envisages the possibility of using small-size trigenerative CAES at the demand

side; ref. [36] presents a trigeneration system combining AA-CAES and heat pump to fulfill cooling, heating and power supply. A cogeneration system combining CAES, organic rankine cycle (ORC) and hybrid compression-absorption refrigeration cycle is proposed in Ref. [37]. The system provides cooling from exhaust hot gas using the ORC driven refrigeration system. Ref. [38] investigates the performance of AA-CAES-based combined cooling, heating, and power generation under different operating strategies in four scenarios. A parametric analysis is conducted to evaluate the influence of key parameters on system performance. Compared with the conventional AA-CAES, a modified system is proposed to take advantage of the heat of discharged air from the low-pressure turbine in Ref. [39]. Ref. [40] reviews a variety of systems that are either integrating CAES technology to a cogeneration system or the CAES system itself operating as a cogeneration system, getting the conclusion that all cogeneration systems with CAES presented good performance. Ref. [41] provides a method of economic assessment for a cogeneration system composed of CAES, organic Rankine cycle and absorption-compression refrigeration cycle; and ref. [42] proposes an AA-CAES based cogeneration system to meet the comprehensive energy demands of a latex factory, where four novel energy supply strategies are analyzed and compared. In the above works, the temperature of thermal demand is neglected. In the presence of multiple thermal demands with different temperatures, new model is needed to exploit the opportunity of cascaded use of thermal energy.

This paper considers modeling and dispatching an AA-CAES based energy hub in an industrial park. The energy hub purchases electricity from the distribution system and serves electric load and thermal demands with two different temperatures. The contribution is listed as follows.

- 1) A comprehensive model of the AA-CAES based energy hub with multi-temperature heat demands is proposed. Electric heater and heat pump are employed to produce high-temperature thermal energy and low-temperature thermal energy, respectively. The flow of heat transfer fluid across high-temperature and low-temperature thermal storage tanks are explicitly considered, allowing modeling the cascaded utilization of heat.
- 2) A bi-level optimization model for optimal bidding and scheduling of AA-CAES based energy hub in the day-ahead distribution market is suggested. The upper level is the power purchase and self-scheduling of the energy hub, aiming at minimizing the daily operation cost; the lower level represents the alternating-current optimal power flow based market clearing which determines the electricity transaction price. As the lower-level market clearing problem is non-linear, existing approaches encounter difficulties to solve such a bilevel problem.
- 3) A direct search algorithm to solve the bi-level model is developed. Radial basis function is used to create a surrogate model, which can be solved with lower computational efforts. The surrogate model is updated dynamically when more sampling points in unvisited regions are available. The proposed method solves the market clearing problem via convex relaxation rather than replacing it with optimality condition, hence is scalable to the size of the power grid. The proposed method outperforms Bayesian Optimization (BO) method, the state-of-the-art surrogate method.

The rest of the paper is organized as follows. The detailed model of the AA-CAES based energy hub is depicted in section 2. The bi-level optimization model for bidding and scheduling in the day-ahead market is presented in Section 3. The radial basis function based surrogate optimization algorithm is developed in section 4. Case studies are reported in Section 5. Conclusions are drawn in

section 6.

## 2. Modeling of AA-CAES EH

The sketch and detailed architecture of the AA-CAES based energy hub are shown in Figs. 1 and 2. The AA-CAES consists of a two-stage compressor, a two-stage expander turbine, an air storage tank, a thermal energy storage system, and external heating source including an electric heater and a heat pump. The thermal energy storage system is made up of three storage tanks with rated temperature of 300 °C (high-temperature tank), 120 °C (low-temperature tank) and 25 °C (ordinary-temperature tank). During charging process, the compressors consumes electricity purchased from the distribution system, producing high-pressure air which is stored in a storage tank or salt cavern; the heat generated by the compression process is collected by the heat exchanger and stored in the high-temperature tank. During discharging process, the high-pressure air is firstly reheated and then flows into the expander. The throttle valves are used to adjust the air pressure in the last-stage compressor and first-stage expander. The thermal energy produced by air compression is limited; so the electric heater and the heat pump are employed to directly heat the high-temperature tank and low-temperature tank, which serve high-temperature demand and low-temperature demand, respectively.

The following assumptions are made in this paper.

- 1) In the air storage tank, the mass and pressure follows ideal gas equation.
- 2) The heat capacities of air and heat transfer fluid are constant.
- 3) The temperature in the air storage tank is constant which is equal to the ambient temperature. The temperatures in the three heat tanks are constant.
- 4) Power consumption of fluid booster pump is neglected. Air pressure loss and heat loss are neglected.

### 2.1. Compressor

According to the adiabatic compression equation of compressor [43,44], the power demand of the  $i$ -th stage is

$$P_{i,t}^c = \dot{m}_t c_p \tau_{i,t}^{c,in} \left( (\beta_i^c)^{\frac{\kappa-1}{\kappa}} - 1 \right) / \eta_i^c \quad (1)$$

where  $\beta_i^c = pr_{i,t}^{c,out} / pr_{i,t}^{c,in}$  is the compression ratio, and  $pr_{i,t}^{c,out} / pr_{i,t}^{c,in}$  is the outlet/inlet air pressure of  $i$ -th stage compressor at  $t$ ;  $\kappa = 1.4$

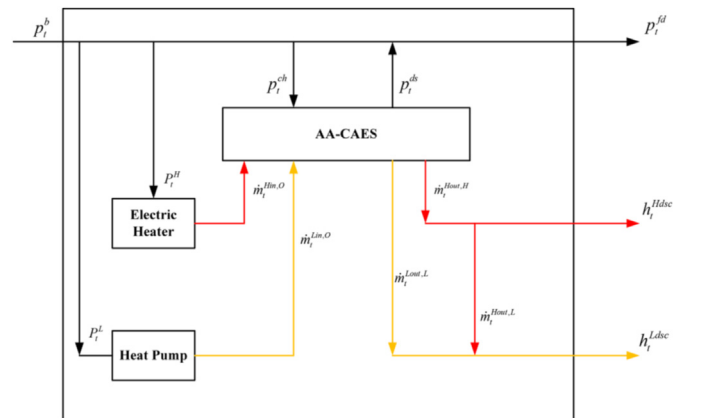


Fig. 1. Sketch of energy hub.

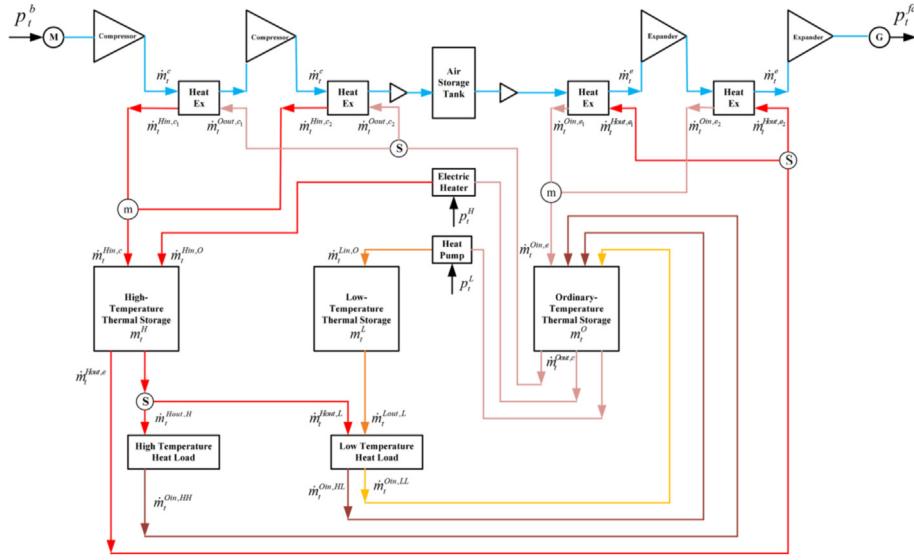


Fig. 2. AA-CAES-based energy hub.

is the adiabatic exponent of air. Hence, the total compressed power demand is

$$P_t^{ch} = \sum_{i=1}^{n_c} P_{i,t}^c \quad (2)$$

The mass flow rate of air through each compression stage meets

$$u_t^c \dot{m}^{c,l} \leq \dot{m}_t^c \leq u_t^c \dot{m}^{c,u} \quad (3)$$

## 2.2. Turbine

In the  $j$ -th stage expander, the released power is

$$P_{j,t}^e = \eta_j^e \dot{m}_t^e c_p^a \tau_{j,t}^{e,in} \left( 1 - \left( \beta_j^e \right)^{\frac{1-k}{k}} \right) \quad (4)$$

The total power generated by the turbine generator is

$$P_t^{ds} = \sum_{j=1}^{n_e} P_{j,t}^e \quad (5)$$

The air mass flow rate in the expansion stage meets

$$u_t^e \dot{m}^{e,l} \leq \dot{m}_t^e \leq u_t^e \dot{m}^{e,u} \quad (6)$$

## 2.3. Air storage tank

In the air storage tank, the temperature and volume are constant; the pressure and mass obeys the ideal gas equation. So the pressure difference in two consecutive periods depends on the change of mass:

$$pr_{t+1}^{st} = pr_t^{st} + \frac{1}{V} R_g \tau^{st} \left( u_t^c \dot{m}_t^c - u_t^e \dot{m}_t^e \right) \quad (7)$$

The air pressure should always reside in the secure range

$$pr^{st,l} \leq pr_t^{st} \leq pr^{st,u} \quad (8)$$

## 2.4. Thermal energy storage system

Heat transfer must satisfy energy conservation law. At the compression side, for the  $i$ -th stage ( $1 \leq i \leq n_c - 1$ ):

$$c_{oil} \dot{m}_t^{Out,c,i} \left( \tau_{i,t}^{c,ex,out} - \tau_{i,t}^{c,ex,in} \right) = c_p^a \dot{m}_t^c \left( \tau_{i,t}^{c,out} - \tau_{i+1,t}^{c,in} \right) \quad (9)$$

For the last stage ( $i = n_c$ ):

$$c_{oil} \dot{m}_t^{Out,c,n_c} \left( \tau_{n_c,t}^{c,ex,out} - \tau_{n_c,t}^{c,ex,in} \right) = c_p^a \dot{m}_t^c \left( \tau_{n_c,t}^{c,out} - \tau_{a,n_c,t}^{c,ex,out} \right) \quad (10)$$

At the expansion side, for the first stage

$$c_{oil} \dot{m}_t^{Hout,e,1} \left( \tau_{1,t}^{e,ex,in} - \tau_{1,t}^{e,ex,out} \right) = c_p^a \dot{m}_t^e \left( \tau_{1,t}^{e,in} - \tau^{st} \right) \quad (11)$$

For the remaining stages ( $2 \leq j \leq n_e$ ):

$$c_{oil} \dot{m}_t^{Hout,e,j} \left( \tau_{j,t}^{e,ex,in} - \tau_{j,t}^{e,ex,out} \right) = c_p^a \dot{m}_t^e \left( \tau_{j,t}^{e,in} - \tau_{j-1,t}^{e,out} \right) \quad (12)$$

According to Fig. 2, the conservation of heat transfer fluid in each heat storage tank can be calculated. For the high-temperature tank, the inflow is from two ways, where one is heat exchange during two-stage air compression, and the other is heating by the electric heater; the outflow is forward to supplying heat for air expansion and providing for high-temperature and low-temperature loads.

$$\dot{m}_{t+1}^H = \dot{m}_t^H + \left( \dot{m}_{t+1}^{Hin,c} + \dot{m}_{t+1}^{Hin,O} \right) - \left( \dot{m}_{t+1}^{Hout,e} + \dot{m}_{t+1}^{Hout,H} + \dot{m}_{t+1}^{Hout,L} \right) \quad (13)$$

For the low-temperature tank, the inflow is only from the heating by the heat pump, and the outflow is for low-temperature load supply.

$$\dot{m}_{t+1}^L = \dot{m}_t^L + \dot{m}_{t+1}^{Lin,O} - \dot{m}_{t+1}^{Lout,L} \quad (14)$$



For the ordinary-temperature tank, the inflow is the outlet side fluid of heat exchangers of air expansion and thermal load supply; the outflow is used for the supplement of high-temperature and low-temperature tanks and heat collection in air compression process.

$$\begin{aligned} m_{t+1}^O = m_t^O - & \left( \dot{m}_{t+1}^{Out,c} + \dot{m}_{t+1}^{Hin,O} + \dot{m}_{t+1}^{Lin,O} \right) \\ & + \left( \dot{m}_{t+1}^{Oin,e} + \dot{m}_{t+1}^{Oin,HH} + \dot{m}_{t+1}^{Oin,HL} + \dot{m}_{t+1}^{Oin,LL} \right) \end{aligned} \quad (15)$$

Heat transfer fluid should comply with mass conservation during air compression and expansion. For the high-temperature tank, the total input fluid from air compression equals the sum of each stage generation; and the total output fluid for air expansion is the sum of each stage usage.

$$\dot{m}_t^{Hin,c} = \dot{m}_t^{Hin,c_1} + \dot{m}_t^{Hin,c_2}, \quad \dot{m}_t^{Hout,e} = \dot{m}_t^{Hout,e_1} + \dot{m}_t^{Hout,e_2} \quad (16)$$

Similarly, for the ordinary-temperature tank, the total output mass for air compression amounts to the sum of fluid consumed in each stage. And the total input mass applied for air expansion equals the sum of each turbine stage.

$$\dot{m}_t^{Oout,c} = \dot{m}_t^{Oout,c_1} + \dot{m}_t^{Oout,c_2}, \quad \dot{m}_t^{Oin,e} = \dot{m}_t^{Oin,e_1} + \dot{m}_t^{Oin,e_2} \quad (17)$$

During air compression, the total and each stage inflow rates of the high-temperature tank equal the total and each stage outflow rates of the ordinary-temperature tank.

$$\dot{m}_t^{Hin,c} = \dot{m}_t^{Oout,c}, \quad \dot{m}_t^{Hin,c_1} = \dot{m}_t^{Oout,c_1}, \quad \dot{m}_t^{Hin,c_2} = \dot{m}_t^{Oout,c_2} \quad (18)$$

During air expansion, the total output rate of the high-temperature tank equals the total input rate of the ordinary-temperature tank. And in each expansion stage, the high-temperature output rate correspondingly equals the ordinary input rate.

$$\dot{m}_t^{Hout,e} = \dot{m}_t^{Oin,e}, \quad \dot{m}_t^{Hout,e_1} = \dot{m}_t^{Oin,e_1}, \quad \dot{m}_t^{Hout,e_2} = \dot{m}_t^{Oin,e_2} \quad (19)$$

For the heat exchangers of thermal supply, the inflow rate of each temperature amounts to the corresponding outflow rate.

$$\dot{m}_t^{Oin,HH} = \dot{m}_t^{Hout,H}, \quad \dot{m}_t^{Oin,LL} = \dot{m}_t^{Lout,L}, \quad \dot{m}_t^{Oin,HL} = \dot{m}_t^{Hout,L} \quad (20)$$

Each tank has a certain volume, yielding

$$m_t^{H,l} \leq m_t^H \leq m_t^{H,u}, \quad m_t^{L,l} \leq m_t^L \leq m_t^{L,u}, \quad m_t^{O,l} \leq m_t^O \leq m_t^{O,u} \quad (21)$$

The temperature of fluid in each heat tank is assumed to be constant, so the fluid heated by the electric heater and heat pump must reach respective rated temperature. The consumption of electric heater and heat pump can be expressed by

$$P_t^H = c_{oil} \dot{m}_t^{Hin,O} \Delta \tau^H / \eta^H, \quad \Delta \tau^H = \tau^H - \tau^O \quad (22)$$

$$P_t^L = c_{oil} \dot{m}_t^{Lin,O} \Delta \tau^L / \eta^L, \quad \Delta \tau^L = \tau^L - \tau^O \quad (23)$$

In case that the low-temperature fluid is insufficient, it is allowed to use high-temperature fluid to supply low-temperature demand, as in Fig. 2, but is prohibited whenever it is unnecessary. Such considerations are modeled by the following set of constraints. For the ordinary-temperature tank:

$$0 \leq \dot{m}_t^{Oout,c} \leq m_t^O - m_t^{O,l} \quad (24)$$

$$0 \leq \dot{m}_t^{Hin,O} + \dot{m}_t^{Lin,O} \leq m_t^O - \dot{m}_t^{Oout,c} - m_t^{O,l} \quad (25)$$

The available ordinary-temperature fluid is first utilized for air compression and then providing supplements for the other two tanks. For the low-temperature tank:

$$0 \leq \dot{m}_t^{Lout,L} \leq m_t^L - m_t^{L,l} \quad (26)$$

$$0 \leq m_t^L - \dot{m}_t^{Lout,L} - m_t^{L,l} \leq M_1(1 - z_1) \quad (27)$$

The output fluid used for low-temperature load supply should not exceed the usable range.  $z_1$  is a binary variable and  $M_1$  is a large enough number. The variable  $z_1$  is set 0 if the available fluid is no less than the demand, and is set 1 to activate higher grade thermal support if the fluid is inadequate. For the high-temperature tank:

$$0 \leq \dot{m}_t^{Hout,e} \leq m_t^H - m_t^{H,l} \quad (28)$$

$$0 \leq \dot{m}_t^{Hout,H} \leq m_t^H - \dot{m}_t^{Hout,e} - m_t^{H,l} \quad (29)$$

$$0 \leq \dot{m}_t^{Hout,L} \leq M_1 z_1 \quad (30)$$

$$0 \leq \dot{m}_t^{Hout,L} \leq m_t^H - \dot{m}_t^{Hout,e} - \dot{m}_t^{Hout,H} - m_t^{H,l} \quad (31)$$

The usable high-temperature fluid is used for air expansion in priority, and thermal supply follows. If  $z_1$  in (30) is 0, the cross-grade heating fluid is enforced to 0 since the lower-grade thermal tank has sufficient surplus. If the high-temperature tank receives the signal of  $z_1 = 1$ , cross-grade heating is active and the supply cannot exceed the rest after satisfying air expansion and high-temperature load requirements.

## 2.5. Startup cost of AA-CAES

Frequent startup and shutdown jeopardize the life of the compressors and expanders. We introduce startup cost  $S^U$ , which can be incorporated in the operation model through

$$SU_t^c \geq 0, \quad SU_t^c \geq S^U(u_t^c - u_{t-1}^c) \quad (32)$$

$$SU_t^e \geq 0, \quad SU_t^e \geq S^U(u_t^e - u_{t-1}^e) \quad (33)$$

From (32), as the cost is to be minimized during compression,  $SU_t^c = S^U$  holds if and only if  $u_t^c = 1$  and  $u_{t-1}^c = 0$ ; for the other values of  $u_t^c$  and  $u_{t-1}^c$ , we must have  $SU_t^c = 0$ . The same holds for the startup cost of expansion in (33).

## 3. Optimal bidding strategy in the distribution system

The AA-CAES based energy hub connects to a node in the distribution system. It bids the power purchase plan  $p_t^b$  in each time period to the distribution power market. Then the market is cleared, and the demand is paid at the nodal price  $\lambda_t$ . The energy hub conducts self-dispatch and meets the local electricity and heat demands, aiming to minimize the daily operation cost. The holistic framework of the bidding and scheduling problem is shown in Fig. 3, where PDN and OPF stand for power distribution network and optimal power flow, respectively.

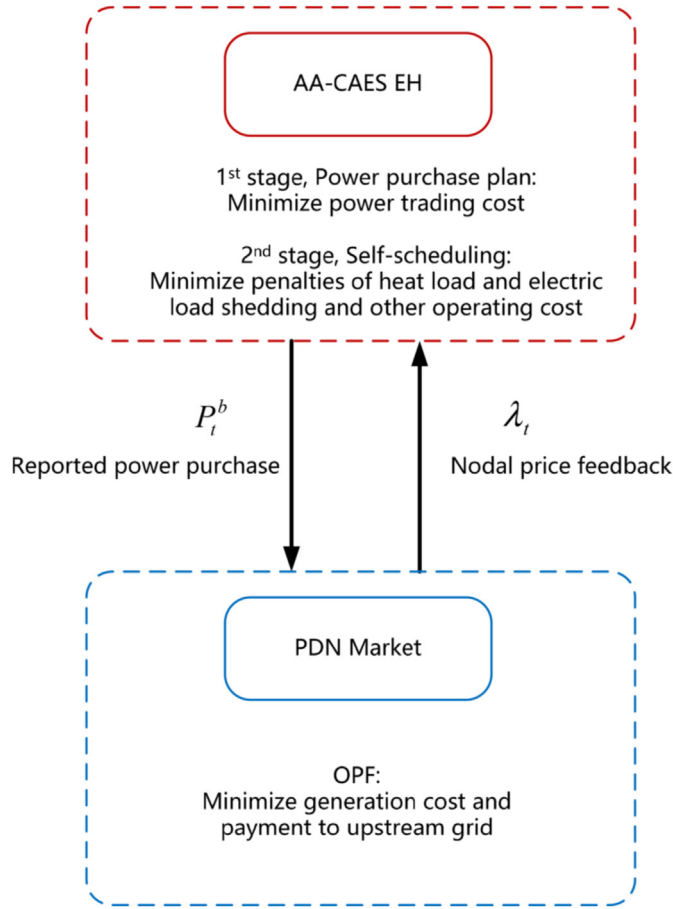


Fig. 3. Framework of optimal dispatch of AA-CAES EH in PDN day-ahead market.

### 3.1. Upper-level problem

The upper level represents the self-scheduling problem of the energy hub. According to Fig. 2, the high-temperature demand is met by the high-temperature fluid, while the low-temperature demand is served by the low-temperature fluid with higher priority and possibly high-temperature fluid if necessary, yielding

$$h_t^{Hdsc} = c_{oil} \dot{m}_t^{Hout,H} \Delta \tau^{Hd} \quad (34)$$

$$h_t^{Ldsc} = c_{oil} \left( \dot{m}_t^{Lout,L} \Delta \tau^{Ld} + \dot{m}_t^{Hout,L} \Delta \tau^{HLd} \right) \quad (35)$$

Load shedding is allowed but will be penalized in the objective function. Load shedding is calculated as

$$h_t^{Hde} = h_t^{Hd} - h_t^{Hdsc}, \quad h_t^{Hde} \geq 0 \quad (36)$$

$$h_t^{Lde} = h_t^{Ld} - h_t^{Ldsc}, \quad h_t^{Lde} \geq 0 \quad (37)$$

For the power demand, the electric power supply is

$$p_t^{dsc} = p_t^b - p_t^H - p_t^L - p_t^{ch} + p_t^{ds}, \quad p_t^{dsc} \geq 0 \quad (38)$$

The load shedding complies with

$$p_t^{de} \geq p_t^{fd} - p_t^{dsc}, \quad p_t^{de} \geq 0 \quad (39)$$

The objective of the energy hub is to minimize the daily

operating cost, including the power purchase cost, startup cost of AA-CAES, and the load shedding penalty:

$$obj = \sum_t \left( \lambda_t p_t^b + SU_t^c + SU_t^e + k_1 h_t^{Hde} + k_2 h_t^{Lde} + k_3 p_t^{de} + k_4 p_t^{over} \right) \quad (40)$$

where  $k_1, k_2, k_3$  are the penalty coefficient for the unit unapplied high-temperature heat load, low-temperature heat load, and electric load, respectively;  $\lambda_t$  is the nodal price which is determined from the lower level market clearing problem. To prevent the energy hub from procuring excessive electricity, penalty for power purchase waste is also added as  $p_t^{over}$  with penalty coefficient  $k_4$ , and  $p_t^{over}$  should satisfy the following constraint:

$$p_t^{over} \geq p_t^{dsc} - p_t^{fd}, \quad p_t^{over} \geq 0 \quad (41)$$

Although the compression/expansion ratio, every inlet/outlet temperature of heat exchangers and mass flow rates are variable, according to Refs. [43,45], an AA-CAES EH often operates at a constant pressure and constant temperature mode in practice, meaning that the power demand/generation of each compression/expansion stage is only associated with the air mass flow rate and thus constraint (1)/(4) is linear. Since the inlet and outlet temperatures of all the heat exchangers work as constants, constraints (9)–(12) are only linear correlation to the mass flow rates of heat transfer fluid and air, and thus (9)–(12) are linear.

### 3.2. Clearance of the distribution market

In the existing research, DC-OPF is the most mature market clearing model. Nevertheless, it assumes constant voltage magnitude and neglects reactive power, and is mainly suitable for lossless transmission networks. However, the AA-CAES energy hub connects to a distribution system, where network loss cannot be neglected, and nodal voltage is an important concern for distribution system operation. Therefore, we use alternating-current (AC) power flow based market clearing model.

Distribution system is operated with radial topology. The typical connection of a distribution system is shown in Fig. 4. The AC power flow can be described by the distflow model [46]:

$$P_{ij}^l + p_j^g - r_{ij}^l I_{ij}^l - p_j^d = \sum_{k \in \pi(j)} P_{jk}^l : \lambda_j \quad (42)$$

$$Q_{ij}^l + q_j^g - x_{ij}^l I_{ij}^l - q_j^d = \sum_{k \in \pi(j)} Q_{jk}^l \quad (43)$$

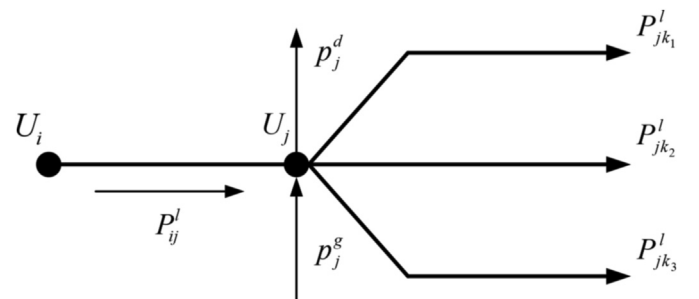


Fig. 4. Typical structure of a radial PDN.

$$U_j = U_i - 2(r_{ij}^l P_{ij}^l + x_{ij}^l Q_{ij}^l) + (z_{ij}^l)^2 I_{ij}^l \quad (44)$$

$$I_{ij}^l U_i = (P_{ij}^l)^2 + (Q_{ij}^l)^2 \quad (45)$$

Equalities (42)–(43) stipulate nodal active and reactive power balancing; nodal price  $\lambda_j$  is the dual variable associated with active balancing condition, indicating the marginal cost of supplying per unit increase of demand at that node. The total active power demand of the bus connected with the energy hub is given by  $p_j^d = p_j^{di} + P^b$ . Voltage drop along distribution line is calculated via (44). The power injection at the head node of each bus is given by (45).

We envision a locational marginal pricing based distribution market [47]. The operator clears the market according to an optimal power flow problem which minimizes the production cost and is cast as

$$\begin{aligned} \min \sum_j [a_j (p_j^g)^2 + b_j (p_j^g)] + \mu \sum_{k \in \pi(0)} p_{0j}^l \\ \text{s.t. (42) – (45)} \end{aligned} \quad (46)$$

lower and upper bounds of variables

In the objective function, the first term is the fuel cost of local generators; in the second term,  $P_{0j}^l$  is the active power in distribution lines connecting to the slack bus, and  $\mu$  is the electricity price at the upstream power grid, so this term is the transaction cost paid to the upstream transmission grid. By solving (46), the dual variable  $\lambda_j$  gives the nodal price which appears in the objective function (40). Since problem (46) is non-convex, we use the convex relaxation method [48]. It replaces the quadratic equality (45) with an inequality  $I_{ij}^l U_i \geq (P_{ij}^l)^2 + (Q_{ij}^l)^2$ , which is actually a second-order cone constraint

$$\begin{bmatrix} 2P_{ij}^l \\ 2Q_{ij}^l \\ U_i - I_{ij}^l \end{bmatrix} \leq U_i + I_{ij}^l \quad (47)$$

So that problem (46) comes down to a second-order cone program. It was shown in Ref. [49] that the convex relaxation in (47) is exact under mild condition, i.e., that inequality naturally holds as equality at the optimal solution of the relaxed problem, so the solution is also the optimal solution of problem (46).

Based on the above discussion, the bidding and scheduling problem of the energy hub can be cast as a bi-level optimization problem

$$\begin{aligned} \min \text{obj} \\ \text{s.t. (1) – (39), (41)} \\ \text{Market clearing problem (46)} \\ \text{in period } t, \quad t = 1 : 24 \end{aligned} \quad (48)$$

In the existing work, to solve a bi-level problem, the lower level is replaced with its KKT optimality condition, in which the complementarity and slackness condition is further linearized via introducing binary variables. However, in problem (48), the lower level is a second-order cone program, its KKT condition involves conic complementarity condition, which cannot be further linearized. The solution algorithm will be developed in the next section.

#### 4. Radial basis function based surrogate optimization algorithm

Problem (48) is difficult to be solved by standard method. Nevertheless, it has a decomposition structure: When the bidding strategy  $\{p_t^b\}_{t=1}^{24}$  is fixed, the upper level reduces to a mixed-integer linear program, and the nodal price can be obtained from the lower level market clearing problem as a second-order cone program. So the cost function of the energy hub is easy to evaluate. The key idea of the proposed method is to create a surrogate model to approximate the objective function in the bidding strategy variable  $\{p_t^b\}_{t=1}^{24}$ .

Surrogate optimization method is popular for black-box optimization problem without derivative information, which means it can only evaluate the function value at discrete points. With the known function values, a surrogate model is built to approximate the objective function; by optimizing such a model, a new sampling point is obtained. The procedure continues until a certain criterion is met [50–52].

However, optimizing with only the surrogate function often converges to a local optimum. Therefore, additional terms are needed to help explore unvisited region. The sum of the surrogate function and the additional terms is minimized to generate the new sampling point [53]. Such a compromised objective makes a tradeoff between searching a new area and the area that may contain a minimum.

We use the renowned radial basis function (RBF) to create the surrogate function. Let  $d(x, x_i) = \|x, x_i\|_2$  denote the Euclidean distance function. The inverse multiquadratic function [50].

$$\psi(\omega d) = \frac{1}{\sqrt{1 + (\omega d)^2}} \quad (49)$$

is employed, where  $\omega > 0$  is a scalar parameter. When  $d = 0$ , and the function takes the maximum 1; when  $d > 0$ , the function value decreases. Hence,  $\Psi(\omega d)$  can be regarded as a spatial weight coefficient. The surrogate function can be constructed as

$$\hat{F}(x) = \sum_{i=1}^N \sigma_i \psi(\omega d(x, x_i)) + \xi_0 \quad (50)$$

where  $\sigma_i$  is the coefficient and  $\xi_0$  is an offset to be calibrated;  $N$  is the number of sampling. According to Ref. [50], the surrogate function should meet the following condition:

$$\hat{F}(x_i) = F_i, \quad i = 1, 2, \dots, N \quad (51)$$

$$\sum_{i=1}^N \sigma_i = 0 \quad (52)$$

where (51) prescribes that the function value at sampling point  $x_i$  must coincide with the value  $F_i$  evaluated from the real model; (52) presents that the sum of the coefficient  $\sigma_i$  is 0.

Solving coefficients  $\sigma_i$  and  $\xi_0$  from equations (50)–(52) we obtain

$$\sigma = M^{-1}(\mathbf{F} - \xi_0 \mathbf{1}) \in \mathbb{R}^{n \times 1} \quad (53)$$

$$M = \psi(\omega \|x_i, x_j\|) \in \mathbb{R}^{n \times n} \quad (54)$$



$$\xi_0 = \left( \mathbf{1}^T M^{-1} \mathbf{1} \right)^{-1} \mathbf{1}^T M^{-1} \mathbf{F} \in \mathbb{R}^{n \times 1} \quad (55)$$

Then, the surrogate function can be obtained as

$$\hat{F}(x) = \Psi^T \sigma + \xi_0 \quad (56)$$

where  $\Psi = [\Psi(\omega \| x_i, x)]_i \in \mathbb{R}^{n \times 1}$ .

We also employ two extra terms to account for the unvisited area in searching the new sampling point. The first one serves as an estimate of the confidence interval associated with the interpolant function. As suggested in Ref. [54] for inverse distance weighting functions, the confidence interval function is considered as

$$p(x) = \sqrt{\sum_{i=1}^N v_i(x) (F_i(x) - \hat{F}(x))^2} \quad (57)$$

where

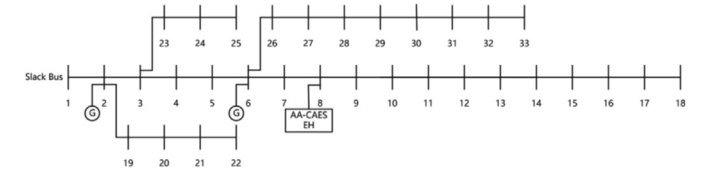


Fig. 5. Typical structure of a radial PDN.

**Table 1**  
Cost coefficients of generators.

| Generator | $a_j$ | $b_j$ |
|-----------|-------|-------|
| Gen-1     | 0.54  | 700   |
| Gen-2     | 0.12  | 900   |

$$A = \hat{F} - \gamma p(x) - \zeta q(x) \quad (61)$$

where  $\gamma$  and  $\zeta$  are non-negative parameters that make balance between minimizing the surrogate function and searching the unvisited region. In the surrogate model (61), the decision variable is  $x = \{p_t^b\}_{t=1}^{24}$ . Flowchart of the surrogate optimization method is summarized in Algorithm 1.

#### Algorithm 1 Surrogate Optimization

- 1: Initiate:  $\gamma \geq 0, \zeta \geq 0, \omega$ , upper and lower bounds  $(\mathbf{l}, \mathbf{u})$ ; the maximum number of function evaluation  $N_{max}$ . Generate  $N_{init}$  random samples  $x_1, \dots, x_{N_{init}}$  with Latin hypercube sampling [55]; set  $N = N_{init}$ .
- 2: Compute the surrogate function  $\hat{F}$  from (50)–(52).
- 3: While  $N < N_{max}$  do  
 Compute the acquisition function (61) and find its minimum  $x_{N+1}$ ;  
 $N = N + 1$ ; update the surrogate function  $\hat{F}$  with newly obtained  $x_{N+1}$ ;  
 End
- 4: Output the optimal bidding strategy  $x^* = x_{N_{max}}$

$$v_i(x) = \begin{cases} 1 & x = x_i \\ 0 & x = x_j, j \neq i \\ \frac{w_i(x)}{\sum_{j=1}^N w_j(x)} & \text{otherwise} \end{cases} \quad (58)$$

and

$$w_i(x) = \frac{e^{-d^2(x, x_i)}}{d^2(x, x_i)} \quad (59)$$

where  $F_i$  denotes the function value at sampling point  $x_i$ .  $p(x)$  promotes areas in which  $\hat{F}$  is more uncertain.

The second one is the inverse distance weighting introduced in Ref. [53].

$$q(x) = \begin{cases} 0 & x \in x_1, \dots, x_N \\ \frac{2}{\pi} \tan^{-1} \left( \frac{1}{\sum_{i=1}^N w_i(x)} \right) & \text{otherwise} \end{cases} \quad (60)$$

which means that  $q(x)$  is zero at sampled points and increases in between. Such terms benefit searching areas without samples.

Finally, the acquisition function is defined as

In Algorithm 1, the market clearing problem is solved only at sampled points via convex relaxation, which gives rise to a second-order cone program. So the proposed method is scalable to the size of power system. The upper-level problem is the self-dispatch of AA-CAES energy hub which is independent of the power system and contains only local decision variables, hence can be readily solved without scalability issue.

## 5. Case study

### 5.1. System settings

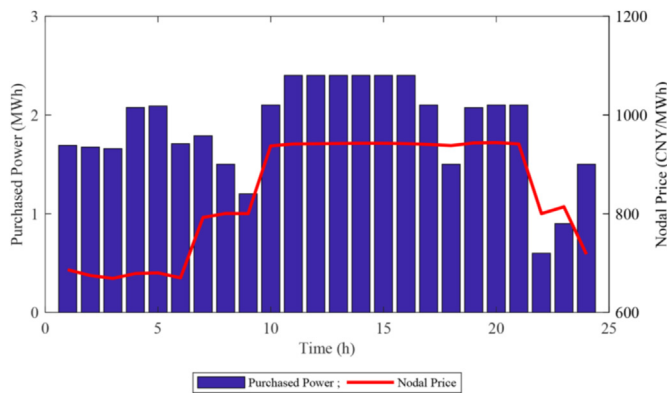
A modified IEEE-33 power system is used to test the proposed method. System topology is shown in Fig. 5. Two generators with rated capacity of 5 MW connect to the system at bus 2 and bus 6, respectively; cost coefficients of generators are listed in Table 1. The peak and valley demands except for the energy hub are 6.3 MW and 1.5 MW. The rated capacities of AA-CAES, the electric heater and the heat pump are 1 MW, 0.9 MW, and 0.35 MW, respectively. Parameters of AA-CAES are given in Table 2 [26]. The electricity price at the slack bus (which stands for the substation) is 1.2 ¥/kWh during peak hours (7 a.m.–9 p.m.) and 0.6 ¥/kWh in the remaining periods. Load shedding penalties are  $k_1 = 5 \times 10^5$  ¥/MWh,  $k_2 = 4.5 \times 10^5$  ¥/MWh and  $k_3 = 10^5$  ¥/MWh. Excessive power purchase penalty is  $k_4 = 2 \times 10^3$  ¥/MWh. Startup cost of  $S^U = 25$  ¥ is used. As the electricity price at the slack bus is time-varying, two

**Table 2**  
Rated parameters of AA-CAES EH.

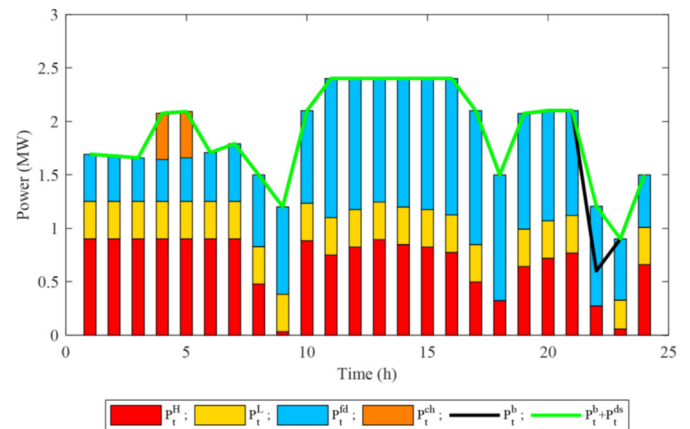
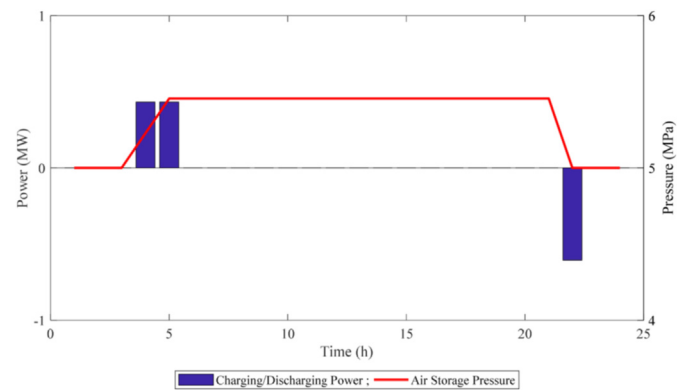
| Compressor                               | 1st-stage | 2nd-stage                         | Turbine                                  | 1st-stage                             | 2nd-stage |
|------------------------------------------|-----------|-----------------------------------|------------------------------------------|---------------------------------------|-----------|
| $\beta_i^c$                              | 10        | 6.5                               | $\beta_j^e$                              | 8.9                                   | 9.4       |
| $\tau_i^{c, in} (^{\circ}\text{C})$      | 25        | 62                                | $\tau_j^{e, in} (^{\circ}\text{C})$      | 264                                   | 264       |
| $\tau_i^{c, out} (^{\circ}\text{C})$     | 337       | 332                               | $\tau_j^{e, out} (^{\circ}\text{C})$     | 62                                    | 58        |
| $\tau_i^{c, ex, in} (^{\circ}\text{C})$  | 25        | 25                                | $\tau_j^{e, ex, in} (^{\circ}\text{C})$  | 300                                   | 300       |
| $\tau_i^{c, ex, out} (^{\circ}\text{C})$ | 300       | 300                               | $\tau_j^{e, ex, out} (^{\circ}\text{C})$ | 61                                    | 98        |
| $\eta_i^c$                               | 0.89      | 0.88                              | $\eta_j^e$                               | 0.81                                  | 0.81      |
| $P_i^c$ (MW)                             | 0.51      | 0.51                              | $P_j^e$ (MW)                             | 0.51                                  | 0.51      |
| $\dot{m}^c$ (kg/s)                       | 1.597     | 1.597                             | $\dot{m}^e$ (kg/s)                       | 2.419                                 | 2.419     |
| $\eta_i^{c, ex}$                         | 0.88      | 0.89                              | $\eta_j^{e, ex}$                         | 0.87                                  | 0.85      |
| Air storage tank                         |           | Thermal storage tank              |                                          |                                       |           |
| $\tau^{st} (^{\circ}\text{C})$           | 25        | $m^{H,u}/m^{H,l}$ (kg)            | $9 \times 10^4/0$                        | $\Delta\tau^{Hd} (^{\circ}\text{C})$  | 198       |
| Pressure range (MPa)                     | 4–10      | $m^{L,u}/m^{L,l}$ (kg)            | $9 \times 10^4/0$                        | $\Delta\tau^{Hld} (^{\circ}\text{C})$ | 196       |
| $V$ (m <sup>3</sup> )                    | 1000      | $m^{O,u}/m^{O,l}$ (kg)            | $9 \times 10^4/0$                        | $\Delta\tau^{Ld} (^{\circ}\text{C})$  | 70        |
| Electric heater                          |           | Heat pump                         |                                          |                                       |           |
| $\eta^H$                                 | 0.98      | $\eta^L$                          | 3                                        |                                       |           |
| $\Delta\tau^H (^{\circ}\text{C})$        | 275       | $\Delta\tau^L (^{\circ}\text{C})$ | 95                                       |                                       |           |

**Table 3**  
AA-CAES EH day-ahead operating cost.

| Iteration                              | 1000  |
|----------------------------------------|-------|
| Total operating cost (CNY)             | 38125 |
| Power purchase cost (CNY)              | 38075 |
| Penalty of high-temperature load (CNY) | 0     |
| Penalty of low-temperature load (CNY)  | 0     |
| Penalty of electric load (CNY)         | 0     |
| Startup cost (CNY)                     | 50    |

**Fig. 6.** Nodal price and purchased power.

generators are dispatched to supply demands during price peak hours. As the AA-CAES energy hub in an industrial park needs to supply intensive load demands, it should be placed close to the slack bus or a generator to prevent long-distance deliver of power, which induces voltage drop along distribution lines. For this consideration, the AA-CAES energy hub connects to the distribution system at bus 8, close to the generator at bus 6. According to the capacity of the energy hub, the bidding demand  $p_t^b \in [0, 3]$  MW. In Algorithm 1, parameters  $\gamma = 0.58$ ,  $\zeta = 0.01$ ,  $\omega = 0.72$ ,  $N_{max} = 1000$ ,  $N_{init} = 48$  are used. All simulations are conducted on a computer with Intel-i7 CPU and 16 GB memory. Optimization problems are coded in MATLAB with YALMIP interface and MOSEK is applied to solve the market clearing problem (46).

**Fig. 7.** AA-CAES EH self-scheduling strategy.**Fig. 8.** SOC of air storage tank.

## 5.2. Simulation results and analysis

### 5.2.1. AA-CAES EH day-ahead scheduling

We set the maximum iteration number 1000, and present numeric results in Table 3.

From the table, it can be observed that the operating cost comes

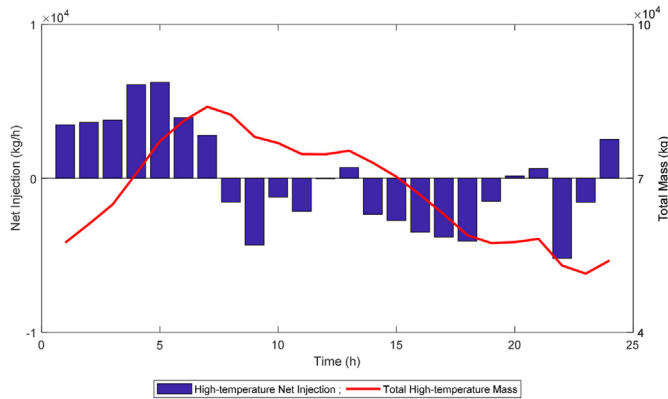


Fig. 9. SOC of high-temperature tank.

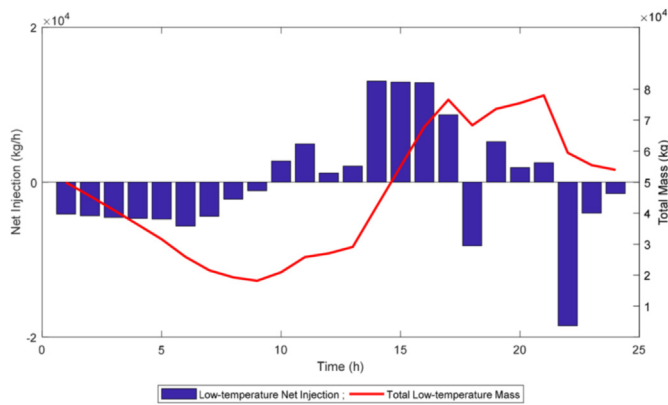


Fig. 10. SOC of low-temperature tank.

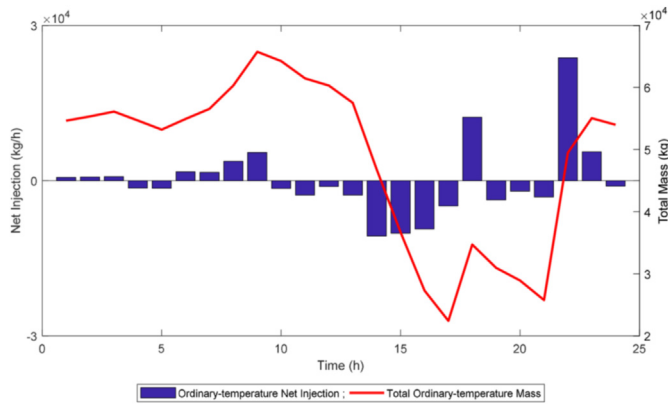


Fig. 11. SOC of ordinary-temperature tank.

from power purchase and AA-CAES startup cost while the penalties of load shedding are zero, indicating that the proposed model can effectively prevent load curtailment and reduce expenditure.

The 24-hour nodal price of bus-8 and the corresponding power procurement are given in Fig. 6.

From the figure, nodal price of bus-8 fluctuates between 688.94 CNY/MWh to 944.01 CNY/MWh and is usually positive correlated to the electric demand. In a typical operating day, the energy hub tends to make average power purchase plan.

Fig. 7 presents the self-scheduling strategy inside AA-CAES EH.

From Fig. 7, due to the constraint of startup cost and charging/

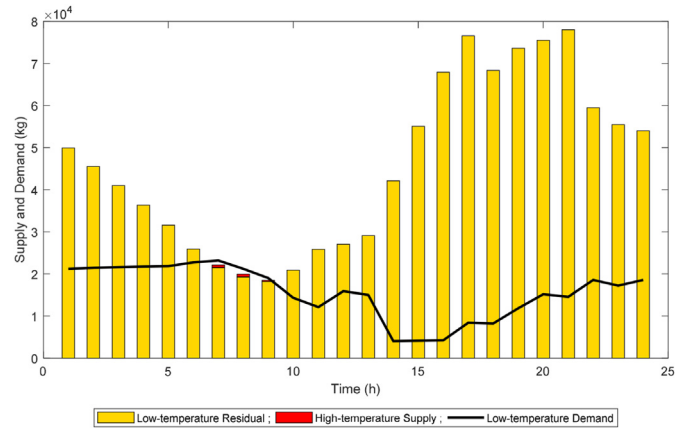


Fig. 12. Cross-grade heating.

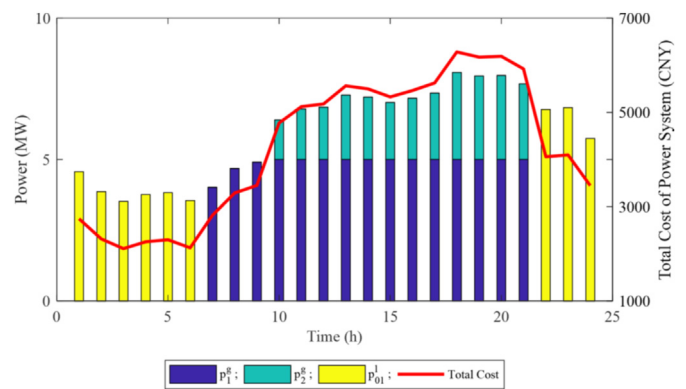


Fig. 13. Dispatch results and operating cost of the distribution power system.

discharging power range, the EH is prone to transfer the excess electricity power into high-temperature and low-temperature thermal energy in priority. Electricity tends to be stored in early morning and be released at late night.

The SOC (State of Charge) of the air storage tank and three thermal storage tanks are plotted in Fig. 8, Fig. 9, Fig. 10 and Fig. 11, respectively.

Electricity is charged in the 4<sup>th</sup> and 5<sup>th</sup> periods and discharged in the 22<sup>nd</sup> period, indicating energy shifting. High-temperature load mainly includes factory production demand like chemical industry, while low-temperature load mostly contains daily general heating and hot water requirement. Since industrial production usually operates during daytime and domestic heating is chiefly needed at night, high-temperature tank tends to store thermal energy at night and largely release in the working hours while the low-temperature tank is prone to supplying heat during night time and central storing in day time periods. The curves in Figs. 9 and 10 match the corresponding working scenarios. The mass of the ordinary-temperature tank fluctuates according to the dynamic of the other two tanks in Fig. 11, indicating the conservation of heat transfer fluid.

### 5.2.2. Result of cross-grade heating

Fig. 12 shows the states of the high-temperature fluid supplying the low-temperature load in the thermal energy storage system.

In most time periods, the remaining available low-temperature fluid is enough for low-temperature demand. Since the low-temperature load in the early morning has been maintained at a

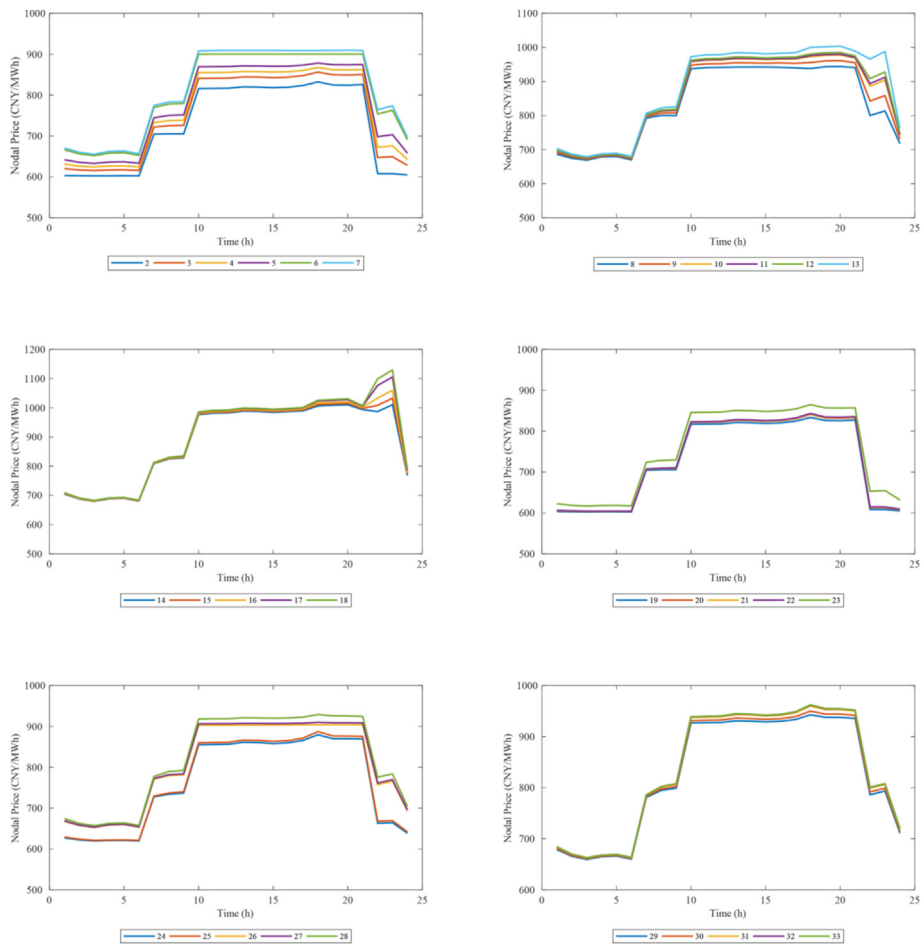


Fig. 14. Nodal prices of node 2 to node 33.

**Table 4**  
Comparison of different algorithms.

| Algorithm   | Iter | Minimum function value (CNY) | Total time cost (s) |
|-------------|------|------------------------------|---------------------|
| Algorithm 1 | 1000 | 38125                        | 1063.2              |
| BO          | 1000 | 40814                        | 1588.1              |

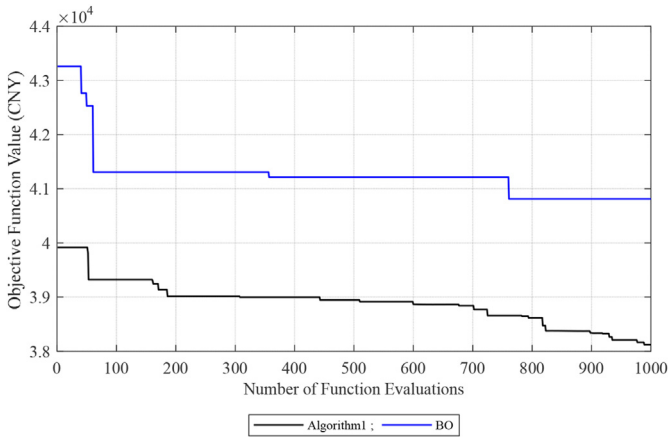


Fig. 15. Comparison of convergence.

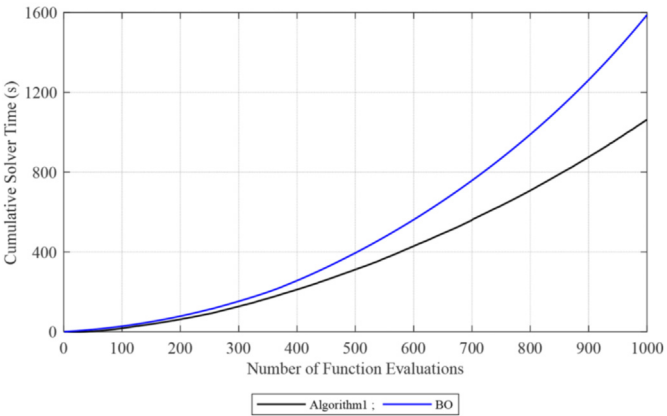


Fig. 16. Comparison of cumulative time consumption.

high level for several hours, the high-temperature fluid centrally supplies the low-temperature demand in  $t_7 - t_9$  to avoid giving up the low-temperature load. The results show that the proposed cross-grade heating model effectively realizes the cascade utilization of thermal energy and promotes the complementation of energy.

**Table 5**

Numerical results of approximate analytical method.

|                                        |       |
|----------------------------------------|-------|
| Total operating cost (CNY)             | 38084 |
| Power purchase cost (CNY)              | 38034 |
| Penalty of high-temperature load (CNY) | 0     |
| Penalty of low-temperature load (CNY)  | 0     |
| Penalty of electric load (CNY)         | 0     |
| Startup cost (CNY)                     | 50    |

### 5.2.3. Results of distribution system operation

The dispatch results and the total operating cost of the distribution power system are presented in Fig. 13, and nodal prices of node 2 to node 33 are shown in Fig. 14, respectively.

From Fig. 13, the power system is prone to procuring power from the slack bus at valley price periods and starting two generators in the middle periods according to the generation cost. From Fig. 14, nodal prices of the whole system at any period vary from 602 CNY/MWh to 1130 CNY/MWh, where the nodes close to the end of the branches have higher prices.

### 5.2.4. Comparison of different optimization algorithms

The same test case is run by different optimization algorithms to illustrate the superiority of the proposed one. To this end, we mainly compare with the computation performance of BO, since BO is a widely used surrogate-based global optimization algorithm [56,57]. The maximum number of function evaluation is set to 1000, which is considered enough to generate relatively usable result in day-ahead scheduling. The final numeric results are recorded in Table 4. And the comparisons of convergence and cumulative time consumption are plotted in Figs. 15 and 16, respectively.

From the table, after 1000 searches, the function value declines to 38125 with the proposed algorithm while only reaches 40814 by BO. From the point of time usage, it takes 1063.2 s for the former while 1588.1 s for the latter to finish the same number of function evaluations. As is shown in Figs. 15 and 16, during the first 1000 iterations, the proposed algorithm possesses better converged performance and higher computation efficiency.

In spite of the advantages, similar to BO, the proposed algorithm cannot judge optimality by a specific condition and can only provide a comparatively ideal solution within limited function evaluations. An analytical method is adopted to verify the accuracy of the proposed method. Since the variable  $P_t^b$  is a parameter in the lower level market-clearing problem, we divide the range of  $P_t^b$  at each period into 30 intervals and use the average value to represent the

nodal price in each interval, thereby obtaining the approximate piece-wise linear (PWL) constant curve of  $\lambda_t(P_t^b)$ .

$$\lambda_t = \begin{cases} A_1, & P_t^b \in [0, 0.1] \\ A_2, & P_t^b \in [0.1, 0.2] \\ \dots & \\ A_{30}, & P_t^b \in [2.9, 3] \end{cases}$$

And  $\lambda_t$  and  $P_t^b$  can be written as the combination of sampling points and variables

$$\lambda_t = \sum_{j=1}^{30} \theta_{jt} A_j, \quad P_t^b = \sum_{j=1}^{31} \varphi_{jt} P_j^b \quad (62)$$

where the binary variables  $\theta_{jt}$  and continuous variables  $\varphi_{jt}$  satisfy the following constraints

$$\varphi_{jt} \geq 0, \quad j = 1, \dots, 31, \quad \sum_{j=1}^{31} \varphi_{jt} = 1 \quad (63)$$

$$\varphi_{1t} \leq \theta_{1t}, \quad \varphi_{31t} \leq \theta_{30t}$$

$$\varphi_{jt} \leq \theta_{j-1t} + \theta_{jt}, \quad j = 2, \dots, 30$$

Substituting the PWL constant of  $\lambda_t(P_t^b)$  to the upper-level objective function and applying further linearization, the original problem can be recast as the following MILP:

$$\min \sum_t \left( \sum_{j=1}^{30} A_j z_{jt} + SU_t^c + SU_t^e + k_1 h_t^{Hde} + k_2 h_t^{Lde} + k_3 P_t^{de} + k_4 P_t^{over} \right)$$

$$\text{s.t. } 0 \leq z_{jt} \leq P^{b,u} \theta_{jt}$$

$$0 \leq P_t^b - z_{jt} \leq P^{b,u} (1 - \theta_{jt})$$

$$(1) - (39), (41), (62) - (63)$$

Market clearing problem (46)

in period  $t$ ,  $t = 1 : 24$

where  $z_{jt}$  is an auxiliary continuous variable equaling  $\theta_{jt} P_t^b$ , and  $P^{b,u}$  is the upper bound of purchase power.

The numerical results computed by the analytical method are shown in Table 5. As the aforementioned calculation results are close to that of the analytical method, the computation accuracy of

**Table 6**

Air temperature gap of heat exchangers.

| Amplification coefficient | 1 <sup>st</sup> -stage compression |             |          | 2 <sup>nd</sup> -stage compression |             |          |
|---------------------------|------------------------------------|-------------|----------|------------------------------------|-------------|----------|
|                           | Inlet (°C)                         | Outlet (°C) | Gap (°C) | Inlet (°C)                         | Outlet (°C) | Gap (°C) |
| 0.85                      | 392                                | 69          | 323      | 392                                | 65          | 327      |
| 0.9                       | 371                                | 67          | 305      | 370                                | 63          | 307      |
| 0.95                      | 353                                | 64          | 289      | 350                                | 61          | 289      |
| 1.0                       | 337                                | 62          | 274      | 332                                | 59          | 273      |
| 1.1                       | 308                                | 59          | 249      | 302                                | 55          | 246      |
| Amplification coefficient | 1 <sup>st</sup> -stage expansion   |             |          | 2 <sup>nd</sup> -stage expansion   |             |          |
|                           | Inlet (°C)                         | Outlet (°C) | Gap (°C) | Inlet (°C)                         | Outlet (°C) | Gap (°C) |
| 0.85                      | 25                                 | 308         | 283      | 122                                | 316         | 194      |
| 0.9                       | 25                                 | 291         | 266      | 100                                | 296         | 196      |
| 0.95                      | 25                                 | 276         | 251      | 80                                 | 279         | 199      |
| 1.0                       | 25                                 | 265         | 240      | 62                                 | 265         | 202      |
| 1.1                       | 25                                 | 241         | 216      | 28                                 | 236         | 208      |



the proposed method is verified. However, the analytical method introduces 792 binary variables, which needs to consume 5,318s to complete optimization. In comparison, the proposed method has more competitive computation efficiency.

The convergence performance of the algorithm depends on the parameters of penalty terms. The parameters  $\gamma$ ,  $\zeta$  and  $\omega$  can be optimized.  $\gamma$  and  $\zeta$  take values between  $[0, 1]$ . The sampling and testing method is adopted to obtain the optimal parameters. The ranges of  $\gamma$  and  $\zeta$  are divided into 10 intervals, respectively. We can try possible combination in turn to find the optimal one. Once a set of algorithm parameters with satisfactory performances is found, it works well under other system parameters. More details about parameter optimization can be found in Ref. [53].

### 5.3. Sensitivity analysis

We also explore the impact of AA-CAES EH parameters and local load scale on the total operating cost, cross-grade supply and electric energy storage. The effect of heat exchanger efficiency and each stage compressor and turbine efficiency on the total operating expense is shown in Figs. 17 and 18, respectively. The studied benchmark parameters are simultaneously multiplied by the same amplification coefficient.

From Fig. 17, it can be observed that the operating cost decreases with the increase of heat exchanger efficiency and the result is opposite in Fig. 18. Higher heat exchanger efficiency means less thermal loss and can thus reduce power procurement cost. However, as is shown in Table 6, higher compressor and turbine efficiency results in lower inlet and outlet air temperature gap of heat exchangers, causing less thermal conversion and storage during compression stage and less heat absorption in air expansion process. Since the efficiency of the two compressors increases more, the reduction of compression heat storage is more than the decrease in expansion heat requirement. Therefore, more electricity needs procuring and converting to heat generation.

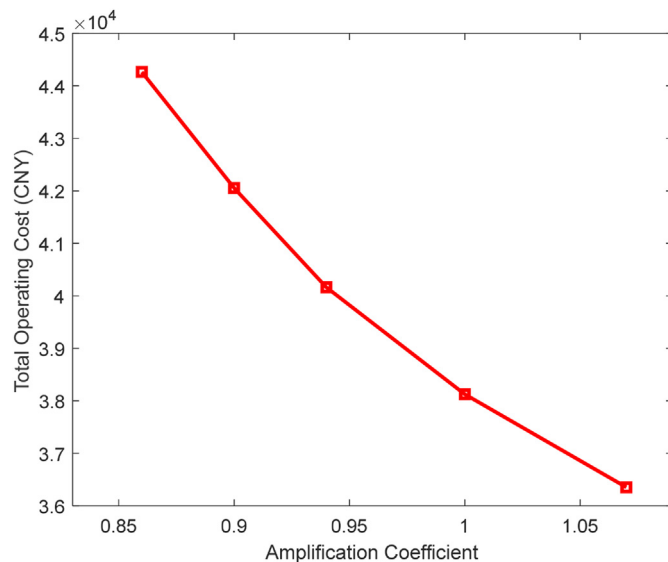


Fig. 17. Impact of heat exchanger efficiency on total operating cost.

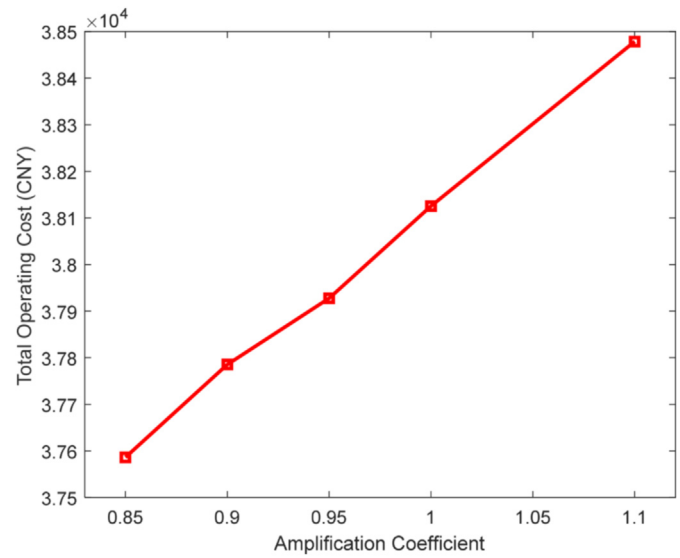


Fig. 18. Impact of compressor and turbine efficiency on total operating cost.

Figs. 19 and 20 present the variation of the total operating cost with changes in thermal and electric loads. Moreover, cross-grade supply curves under different levels of thermal load are plotted in Fig. 21 and air storage SOCs at different electric load levels are provided in Fig. 22.

From Figs. 19 and 20, the total operating cost increases with the extending loads. In Fig. 21, high-temperature storage supplies more heat before 10 a.m. when low-temperature demand gradually becomes higher. And Fig. 22 illustrates that the charge and discharge capacity of AA-CAES is not necessarily related to the electric load level. Nonetheless, AA-CAES basically tends to store power in early hours of a day and largely release in later peak hours, realizing efficient energy utilization.

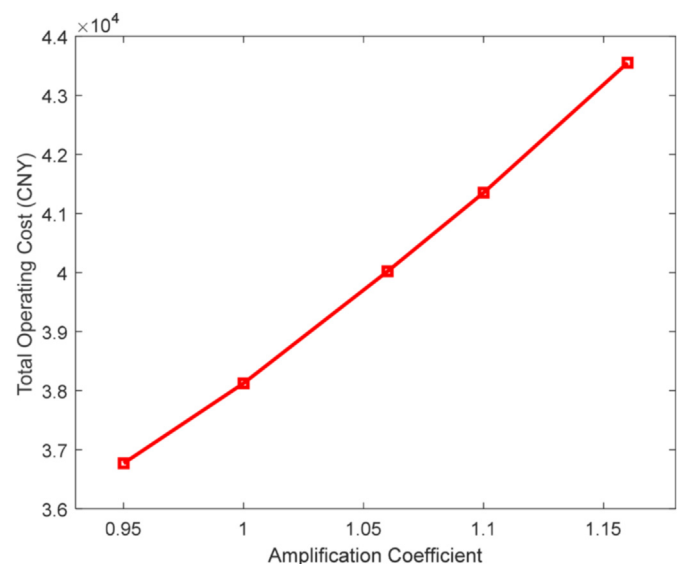


Fig. 19. Impact of thermal load level on total operating cost.

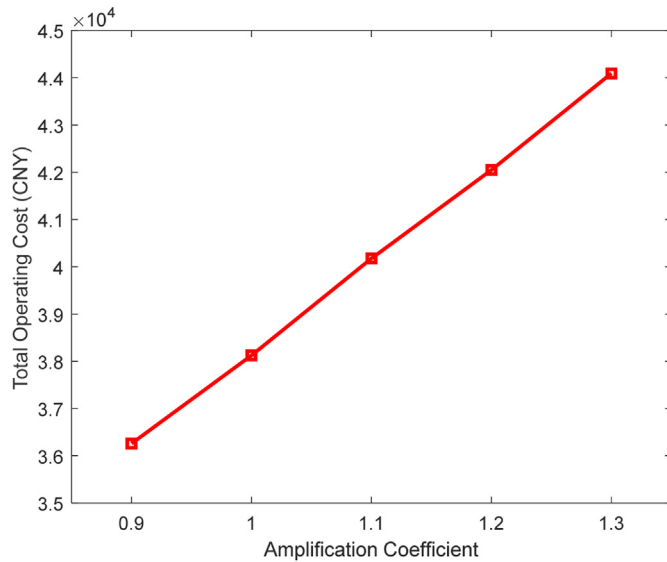


Fig. 20. Impact of electric load level on total operating cost.

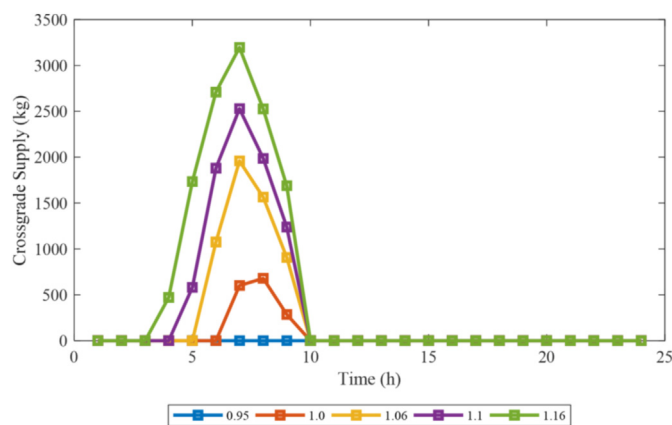


Fig. 21. Cross-grade supply under different thermal load levels.

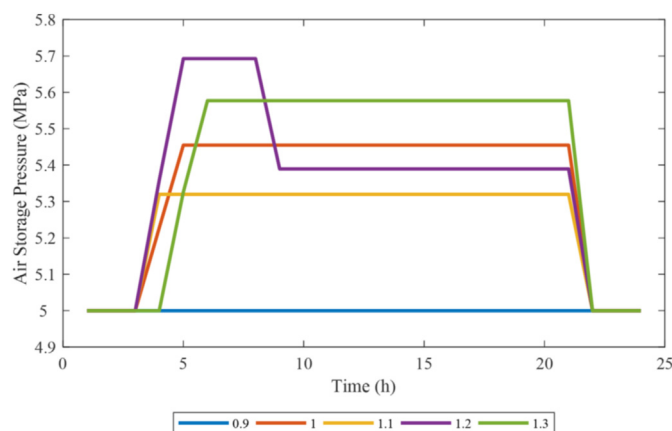


Fig. 22. Air storage SOCs at different electric load levels.

## 6. Conclusions

This paper proposes a comprehensive model for AA-CAES EH day-ahead scheduling of power and heat. The model has a complete depiction of the characteristics in AA-CAES, including air storage system and thermal energy storage system. Based on this, the model fully considers combined heat and power supply and cross-grade heating, realizing high-efficiency use of energy with lowest expenditure. Numeric experiments verify the effectiveness of energy dispatch under cost minimization. Sensitivity analysis is also carried out to explore the impact of AA-CAES EH parameters and local load scale on the total operating cost, cross-grade heat supply and electric power storage.

In addition, an RBF based global optimization algorithm is also put forward to achieve a high-quality solution within limited number of function evaluations. Not only does the algorithm simplify the original function with RBF based surrogate function, but uses two penalty terms to consider the uncertainty near the surrogate and explore the feasible regions that have not yet been visited. From the numeric tests, the proposed algorithm enjoys a better computation performance and efficiency than BO. Nonetheless, two aspects need to pay attention: 1) the algorithm cannot guarantee the optimality, which is same as BO, but can obtain a value which is more acceptable in the day-ahead extent; 2) the descent performance of the algorithm has a strong correlation with the coefficients of the penalty functions, and thus the coefficients need to be optimized first, which may be time-consuming.

## Credit author statement

Danman Wu: Software, Validation, Writing – original draft. Jiayu Bai: Formal analysis, Validation. Wei Wei: Conceptualization, Methodology, Formal analysis, Writing – original draft. Laijun Chen: Writing – review & editing. Shengwei Mei: Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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